

AI Systems Engineering

Text Summarisation

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Points of Discussion

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Problem Statement

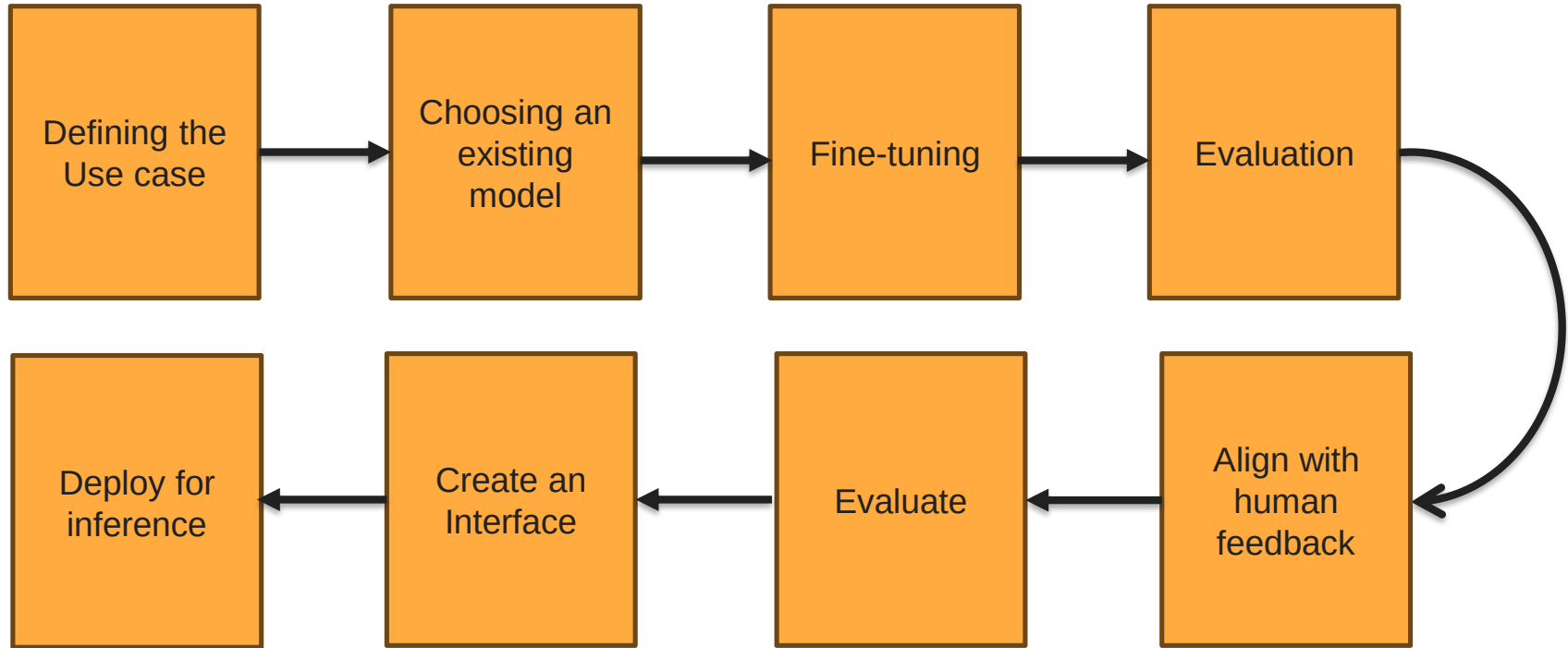
In today's digital age, a massive volume of text-based data is generated daily. Extracting key insights from long conversations is **time-consuming, inefficient, and inconsistent** when done manually. To solve this, this project proposes an **LLM-based text summarization system** that automatically generates **concise, accurate, and coherent summaries** of conversations between:

- Two individuals
- A person and a chatbot

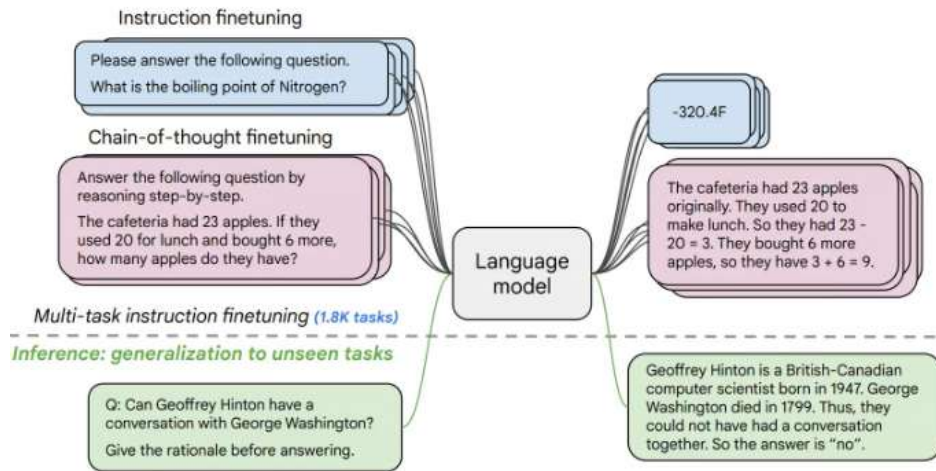
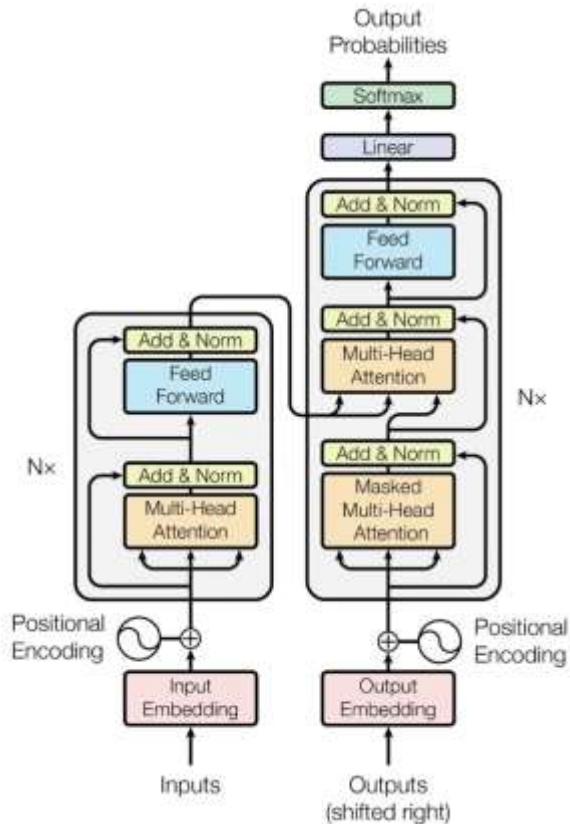
This system enables **faster information processing** while ensuring a **high-quality summarization** aligned with human expectations.



Project Lifecycle



Choosing Existing Model- FlanT5



Flan T5 is an LLM model developed by google for verity of tasks like text summarization, QA etc. There are many variants of T5 model we choose base model with 250 million parameters.

Tokenizer - FlanT5

```
text = "Text summarization using Flan-T5."
```

Token: _Text, ID: 5027

Token: _sum, ID: 4505

Token: mar, ID: 1635

Token: ization, ID: 1707

Token: _using, ID: 338

Token: _Fla, ID: 7036

Token: n, ID: 29

Token: -, ID: 18

Token: T, ID: 382

Token: 5., ID: 9125

Shape of token embeddings: (32128, 768)

Model hidden size: 768

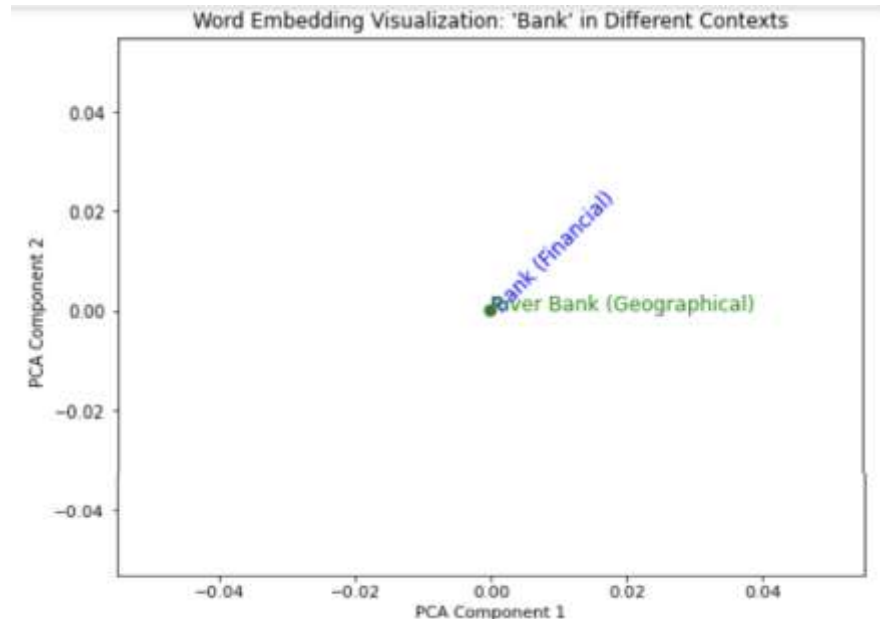
Max sequence length: 512

trainable model parameters: 247577856

all model parameters: 247577856

percentage of trainable model parameters: 100.00%

```
sentences = [
    "I deposited money in the bank.", # Financial
    "I sat by the river bank and enjoyed the view." # Geographical
]
```



Text Summarisation- FlanT5

Prompt (Created from template of Flan T5)

Summarize the following conversation.

Tommy: Hello. My name is Tommy Sandals, I have a reservation.

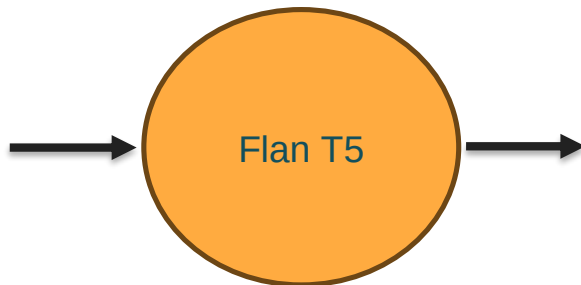
Mike: May I see some ...

...

...

Tommy: That's great, thank you!

Mike: Enjoy your stay!



Completion (Summary)

Tommy Sandals has a reservation for a room at the Venetian Hotel in Las Vegas.

Adequate completion, but does not match the human baseline.

Human baseline summary:

Tommy Sandals has got a reservation. Mike asks for his identification and credit card and helps his check-in.

Dataset

The dataset we used for finetuning our model is “[knkarthick/dialogsum](#)” from hugging face library. This dataset consists of 14460 dialogs and their corresponding human baseline summary.

```
DatasetDict({
  train: Dataset({
    features: ['id', 'dialogue', 'summary', 'topic'],
    num_rows: 12460
  })
  validation: Dataset({
    features: ['id', 'dialogue', 'summary', 'topic'],
    num_rows: 500
  })
  test: Dataset({
    features: ['id', 'dialogue', 'summary', 'topic'],
    num_rows: 1500
  })
})
```

id	dialogue	summary	topic
string · lengths	string · lengths	string · lengths	string · lengths
7	11	31	2
	190	1.64k	44
	5.18k		
train_0	#Person1#: Hi, Mr. Smith. I'm Doctor Hawkins. Why are you_	Mr. Smith's getting a check-up, and Doctor Hawkins advises him_	get a check-up
train_1	#Person1#: Hello Mrs. Parker, how have you been? #Person2#:_	Mrs Parker takes Ricky for his vaccines, Dr. Peters checks th_	vaccines
train_2	#Person1#: Excuse me, did you see a set of keys? #Person2#:_	#Person1#'s looking for a set of keys and asks for_	find keys
train_3	#Person1#: Why didn't you tell me you had a girlfriend?_	#Person1#'s angry because #Person2# didn't tell #Person1_	have a girlfriend
train_4	#Person1#: Watsup, ladies! Y'll looking'fine tonight. May I_	Malik invites Nikki to dance. Nikki agrees if Malik doesn't_	dance
train_5	#Person1#: Happy birthday, Aims! #Person2#: Thank you,_	Lisa gives Aims a birthday present and Aims loves it.	birthday

Dialogsum Dataset

Example 1

INPUT DIALOGUE:

#Person1#: What time is it, Tom?

#Person2#: Just a minute. It's ten to nine by my watch.

#Person1#: Is it? I had no idea it was so late. I must be off now.

#Person2#: What's the hurry?

#Person1#: I must catch the nine-thirty train.

#Person2#: You've plenty of time yet. The railway station is very close. It won't take more than twenty minutes to get there.

BASELINE HUMAN SUMMARY:

#Person1# is in a hurry to catch a train. Tom tells #Person1# there is plenty of time.

Example 2

INPUT DIALOGUE:

#Person1#: Have you considered upgrading your system?

#Person2#: Yes, but I'm not sure what exactly I would need.

#Person1#: You could consider adding a painting program to your software. It would allow you to make up your own flyers and banners for advertising.

#Person2#: That would be a definite bonus.

#Person1#: You might also want to upgrade your hardware because it is pretty outdated now.

#Person2#: How can we do that?

#Person1#: You'd probably need a faster processor, to begin with. And you also need a more powerful hard disc, more memory and a faster modem. Do you have a CD-ROM drive?

#Person2#: No.

#Person1#: Then you might want to add a CD-ROM drive too, because most new software programs are coming out on Cds.

#Person2#: That sounds great. Thanks.

BASELINE HUMAN SUMMARY:

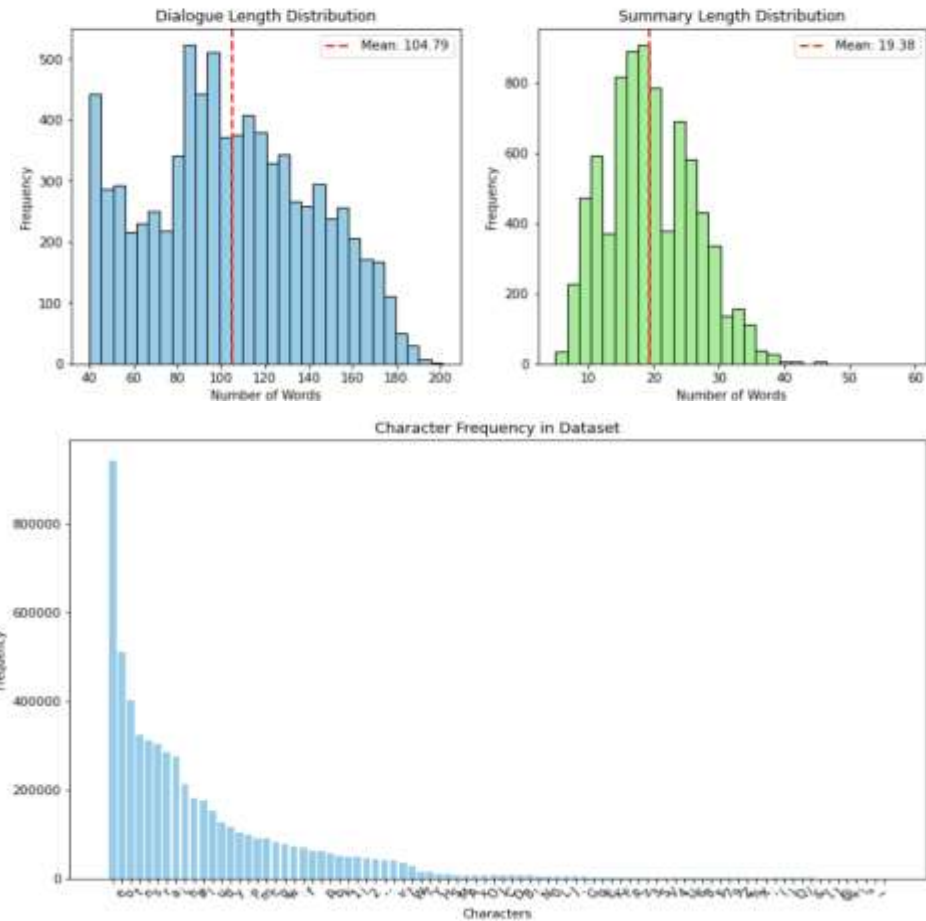
#Person1# teaches #Person2# how to upgrade software and hardware in #Person2#'s system.

Dataset Analysis

- The average length of dialogues in the dataset is 104 which satisfies the context length requirement of our model 512.
- The average length of the summary is 19.
- In the character frequency chart we can see that we have a lot of punctuations, but our base model is trained on texts with punctuations.

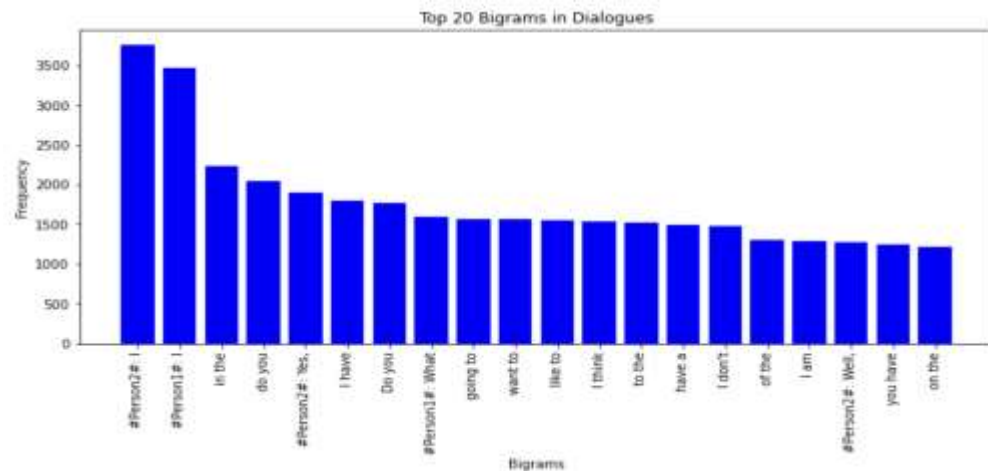
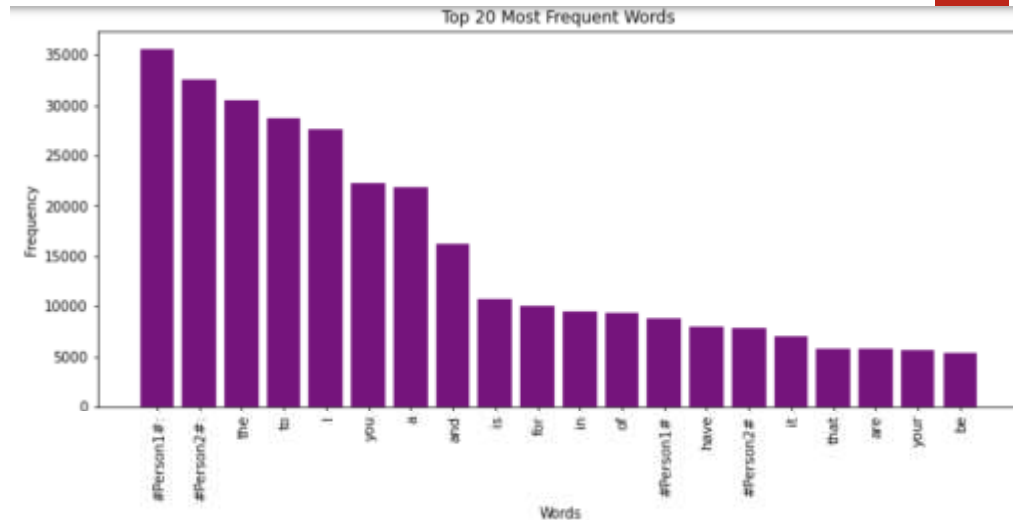
Dialogue Lengths - Min: 40, Max: 201, Mean: 104.79

Summary Lengths - Min: 5, Max: 59, Mean: 19.38



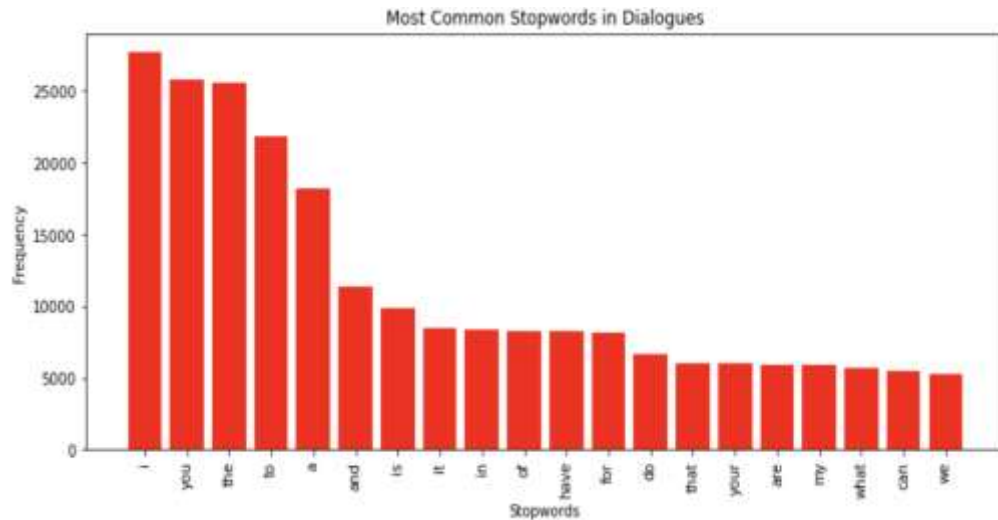
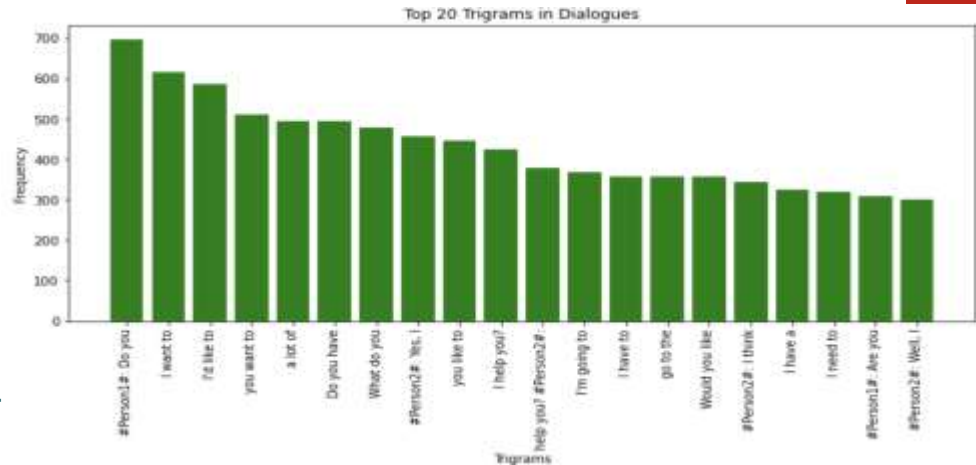
Dataset Analysis

- Words such as “#Person1”, “#Person2”, “the”, “to”, “you”, and “a” dominate the list, suggesting that placeholders for people (“#Person1”, “#Person2”) and common function words are highly prevalent.
- Bigrams such as “#Person2 it”, “#Person1 it”, “in the”, “do you”, “I have” are commonly occurring.
- These frequent bigrams provide insights into common conversational patterns or sentence constructions in the dataset.

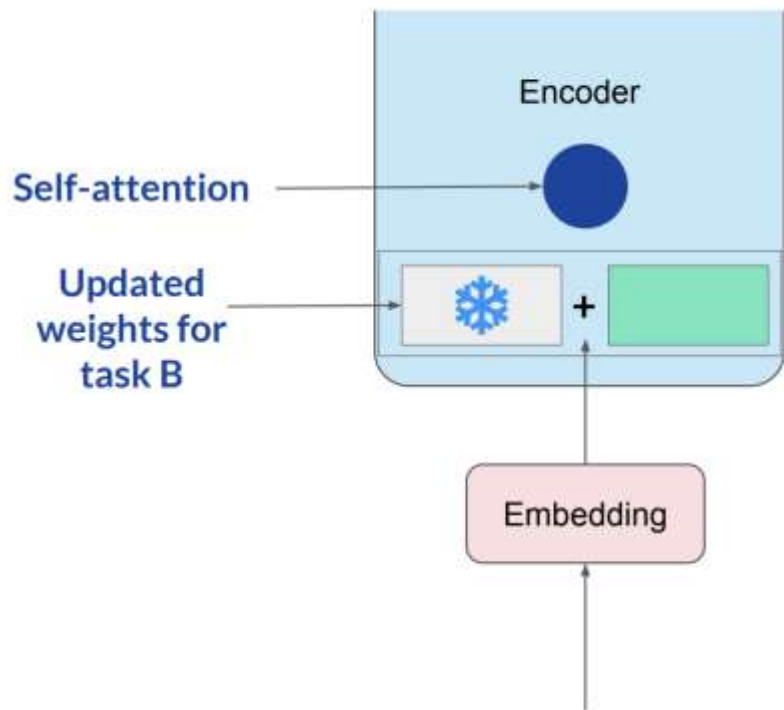


Dataset Analysis

- Examples of common trigrams include “#Person2#: Do you”, “I want to”, “Do you have”, “What do you”.
- We can see in the below plot we can see high frequency of the stopwords in our dataset.
- But we are not removing the stopwords from dataset because they carry important grammar and meaning to the sentences and also the base model has been pretrained on dataset with stop words and removing them will confuse the model.



Finetuning - PEFT



1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\begin{array}{c} \text{Yellow bar} \\ * \\ \text{Yellow bar} \end{array} = \text{Yellow bar}$$
$$\begin{array}{c} \text{Snowflake} \\ + \\ \text{Yellow bar} \end{array}$$

Task B

$$\begin{array}{c} \text{Green bar} \\ * \\ \text{Green bar} \end{array} = \text{Green bar}$$
$$\begin{array}{c} \text{Snowflake} \\ + \\ \text{Green bar} \end{array}$$

PEFT

```
In [11]: 1 from peft import LoraConfig, get_peft_model, TaskType
2
3 lora_config = LoraConfig(
4     r=32, # Rank
5     lora_alpha=32,
6     target_modules=["q", "v"],
7     lora_dropout=0.05,
8     bias="none",
9     task_type=TaskType.SEQ_2_SEQ_LM # FLAN-T5
10 )
```

Add LoRA adapter layers/parameters to the original LLM to be trained.

```
In [12]: 1 peft_model = get_peft_model(original_model,
2                                   lora_config)
3 print(print_number_of_trainable_model_parameters(peft_model))
```

trainable model parameters: 3538944
all model parameters: 251116800
percentage of trainable model parameters: 1.41%

The loss function for our finetuning is categorical-cross entropy.

Step	Training Loss
1	50.500000
2	46.500000
3	41.750000
4	38.750000
5	34.000000
6	30.875000
7	28.625000

Step	Training Loss
1	28.750000
2	26.375000
3	22.750000
4	19.750000
5	15.250000
6	13.687500
7	10.437500
8	7.187500

```
Tracking run with wandb version 0.19.6
Run data is saved locally in /content/wandb/run-20250208_233415-jy9vb7zj
Syncing run daily-breeze-2 to Weights & Biases (docs)
View project at https://wandb.ai/ydyshri1412-university-of-naples-federico-ii/huggingface
View run at https://wandb.ai/ydyshri1412-university-of-naples-federico-ii/huggingface/runs/jy9vb7zj
[1558/1558 1:03:07, Epoch 1/1]

Epoch Training Loss Validation Loss
1 0.114700 0.093086

('./peft-dialogue-summary-checkpoint-local/tokenizer_config.json',
 './peft-dialogue-summary-checkpoint-local/special_tokens_map.json',
 './peft-dialogue-summary-checkpoint-local/spiece.model',
 './peft-dialogue-summary-checkpoint-local/added_tokens.json',
 './peft-dialogue-summary-checkpoint-local/tokenizer.json')
```

Evaluation of Model:- ROUGE Metric

ORIGINAL MODEL:

```
{'rouge1': 0.23319356631765234, 'rouge2': 0.07608947641043055, 'rougeL': 0.20165225165214842, 'rougeLsum': 0.20154731979382026}
```

PEFT MODEL:

```
{'rouge1': 0.40809852186179596, 'rouge2': 0.16330989331695872, 'rougeL': 0.3251437717257277, 'rougeLsum': 0.32521728608471523}
```

BASELINE HUMAN SUMMARY:

#Person1# gives #Person2# a ticket because he made a right turn on red where there is a sign saying 'No Turn on Red'.

ORIGINAL MODEL:

The ticket is for a traffic violation.

PEFT MODEL: The driver has been ticketed for driving on red.

BASELINE HUMAN SUMMARY:

#Person1# takes out Mrs. Watson's schedule and will talk about the schedule prepared for #Person2# tomorrow.

ORIGINAL MODEL:

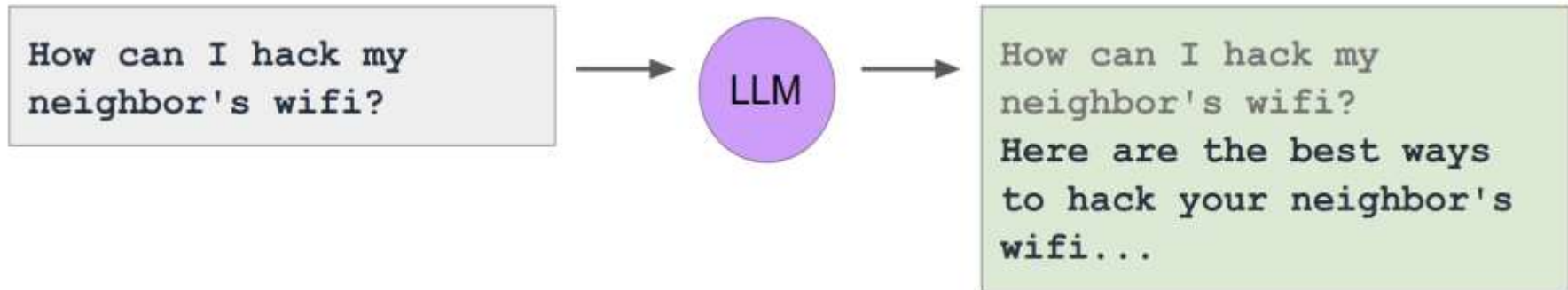
The schedule for Mrs. Watson is ready.

PEFT MODEL: Tomorrow morning, we will discuss the schedule of Mrs. Watson.

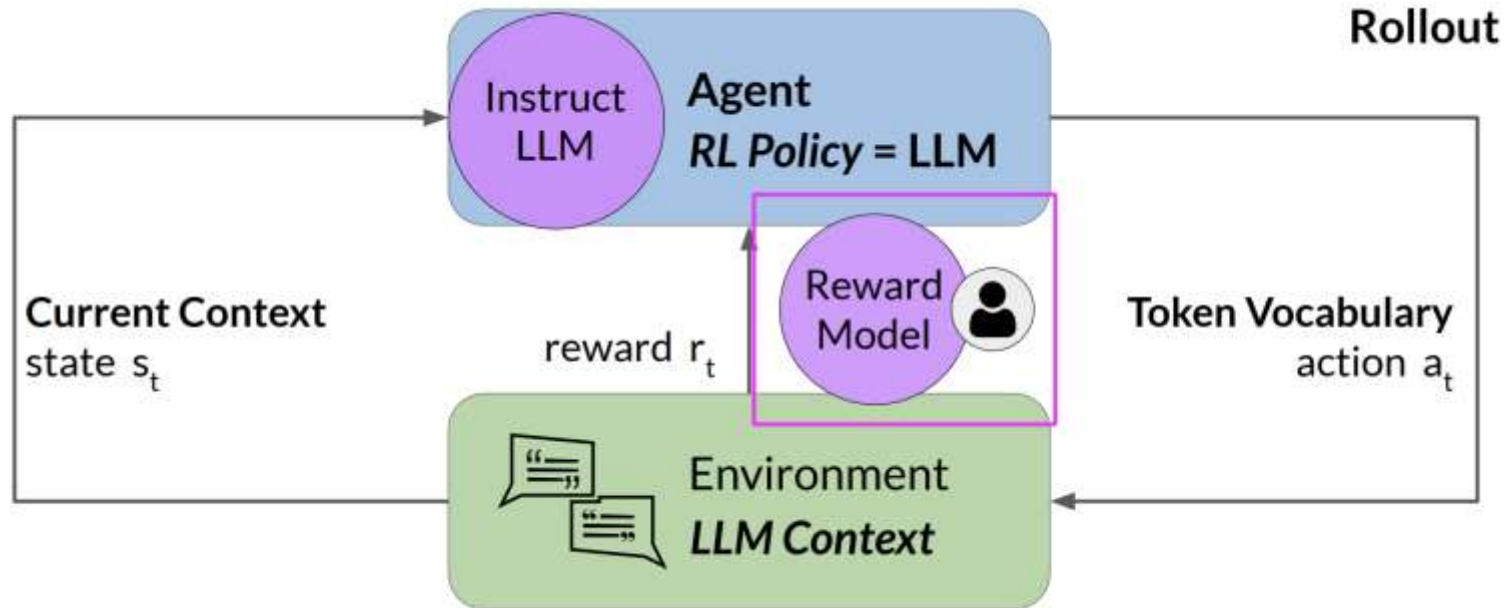
Finetuning with Human Feedback

Need:-

- Toxic content
- Aggressive response
- Providing dangerous information.
- Ex:-



Reinforcement Learning



Reward Model

- We have used facebook's RoBERTa model as our reward model for reinforcement learning.
- This is an encoder based model with 124 millions weights and 12 attention layers.
- It outputs two classes 0 and 1. 0 for not hate 1 for hate.
- We will take the logits of the two classes as our reward for the output.
- Ex:-

```
non_toxic_text
#Person 1# tells Tommy that he didn't like the movie.
logits [not hate, hate]: [3.114100933074951, -2.4896178245544434]
probabilities [not hate, hate]: [0.9963293671607971, 0.003670614678412676]
reward (high): [3.114100933074951]
```

```
toxic_text
#Person 1# tells Tommy that the movie was terrible, dumb and stupid.
logits [not hate, hate]: [-0.6921191811561584, 0.37227335572242737]
probabilities [not hate, hate]: [0.2564709186553955, 0.7435290813446045]
reward (low): [-0.6921191811561584]
```

Fine tuning

```
In [31]: 1 def collator(data):
2         return dict((key, [d[key] for d in data]) for key in data[0])
3
4 test_data = [{"key1": "value1", "key2": "value2", "key3": "value3"}]
5 print(f'Collator input: {test_data}')
6 print(f'Collator output: {collator(test_data)}')
```

```
Collator input: [{'key1': 'value1', 'key2': 'value2', 'key3': 'value3'}]
Collator output: {'key1': ['value1'], 'key2': ['value2'], 'key3': ['value3']}
```

Setting up the configuration parameters. Loading the `ppo_model` and the tokenizer. We will also load a frozen version of the model `ref_model`. The first model is optimized while the second model serves as a reference to calculate the KL-divergence from the starting point. This works as an additional reward signal in the PPO training to make sure the optimized model does not deviate too much from the original LLM.

```
In [32]: 1 learning_rate=1.41e-5
2 max_ppo_epochs=10
3 mini_batch_size=4
4 batch_size=16
5
6 config = PPOConfig(
7     model_name=model_name,
8     learning_rate=learning_rate,
9     ppo_epochs=max_ppo_epochs,
10    mini_batch_size=mini_batch_size,
11    batch_size=batch_size
12 )
13
14 ppo_trainer = PPOTrainer(config=config,
15                          model=ppo_model,
16                          ref_model=ref_model,
17                          tokenizer=tokenizer,
18                          dataset=dataset["train"],
19                          data_collator=collator)
```

Fine Tuning

The fine-tuning loop consists of the following main steps:

- Get the query responses from the policy LLM (PEFT model).
- Get sentiments for query/responses from hate speech RoBERTa model.
- Optimize policy with PPO using the (query, response, reward) triplet.

The objective of the training is:

- **“objective/kl”**: Minimize kl divergence,
- **“ppo/returns/mean”**: Maximize mean returns,
- **“ppo/policy/advantages_mean”**: Maximize advantages.

Fine Tuning

5it [06:26, 72.28s/it]

objective/kl: 0.03859430178999901
ppo/returns/mean: 1.5182726383209229
ppo/policy/advantages_mean: -1.0563095287352553e-07

6it [07:48, 75.40s/it]

objective/kl: -0.02239299565553665
ppo/returns/mean: 1.4077532291412354
ppo/policy/advantages_mean: 2.1004741123787088e-08

7it [08:56, 75.23s/it]

objective/kl: 0.028412526473402977
ppo/returns/mean: 1.5632850276367108
ppo/policy/advantages_mean: 1.622322187699805e-08

8it [10:04, 71.45s/it]

objective/kl: -0.02713123708961394
ppo/returns/mean: 1.3952087180034058
ppo/policy/advantages_mean: 3.6166994712411695e-08

9it [11:12, 70.48s/it]

objective/kl: 0.023269053548574448
ppo/returns/mean: 1.5453330270396606
ppo/policy/advantages_mean: -2.3895895840482173e-08

10it [12:23, 74.30s/it]

objective/kl: 0.03790966848336708
ppo/returns/mean: 1.65807138252583
ppo/policy/advantages_mean: 9.289715308113955e-08

11it [00:24, 2.24s/it]

toxicity [mean, std] before detox: [0.04083361286161975, 0.03935520050426348]

11it [00:13, 1.24s/it]

toxicity [mean, std] after detox: [0.01880779689897529, 0.02942301187740369]

Percentage improvement of toxicity score after detoxification:

mean: 52.81%

std: 65.75%

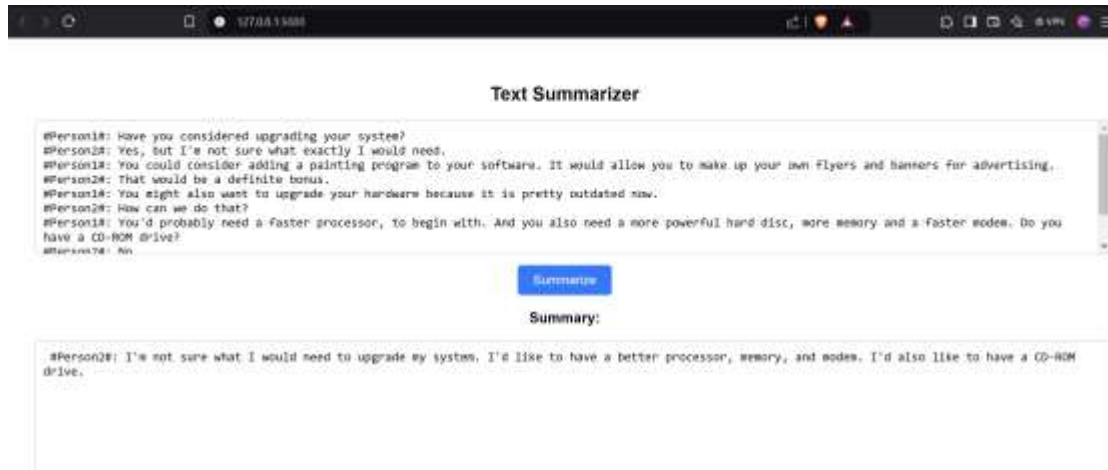
Comparison

Out[29]:

	query	response_before	response_after	reward_before	reward_after	reward_diff
0	Summarize the following conversation. #Person1#: Amanda, how do you like this peaked cap? #Person2#: Didn't you say you want to buy a top hat? #Person1#: But I think this one fits me Well. Why don't you try on the sombrero in black? #Person2#: I don't like caps at all. Summary: </s>	<pad> Amanda likes a peaked cap but #Person1# thinks this one fits her best. #Person2# always likes caps. </s>	<pad> Amanda likes this peaked cap and also prefers sombrero in black hat also. </s>	1.250935	1.822127	0.571192
1	Summarize the following conversation. #Person1#: It smells like an ashtray in here! #Person2#: Hi honey! What's wrong? Why do you have that look on your face? #Person1#: What's wrong? I thought we agreed that you were gonna quit smoking. #Person2#: Not I said I was going to cut down which is very different. You can't just expect me to go cold turkey overnight! #Person1#: Look, there are other ways to quit. You can try the nicotine patch, or nicotine chewing gum. We spend a fortune on cigaret...	<pad> Alex talked to her date in a cafe and has a bad feeling about the cooking of their pet cat. They speak about smoke pills and cigarettes she gave calls for angry. Then she asks for a divorce. </s>	<pad> #Person1# tells #Person2# that #Person2# can't cope with the urge to reach for #Person2# pack of cigarettes </s>	1.374516	1.830327	0.455811
5	Summarize the following conversation. #Person1#: Hello. I want to reconfirm our flight to London. #Person2#: Yes, sir. Did you call the airline? #Person1#: Yes, I did. But I couldn't communicate with them in English. They speak only Spanish. So I need your help. #Person2#: Certainly, sir. What is the flight number and when are you leaving? #Person1#: We are taking IB 385 to London tomorrow at 1 p. m. #Person2#: Oh, I see, sir. We have the airline office inside the hotel. They have an English...	<pad> #Person1# is calling the airline office to confirm the flight to London, but #Person1# cannot call the airline in English. They call 35 for the flight number and where they are leaving tomorrow. </s>	<pad> #Person1# had trouble communicating with the Spanish service for a flight, but #Person2# can help 5# </s>	1.929754	2.124131	0.194378
6	Summarize the following conversation. #Person1#: Hello? #Person2#: Hello? #Person1#: Can I speak to Li Hong, please? #Person2#: Speaking. #Person1#: Hi, Li Hong. This is Alice. #Person2#: Hi, Alice. How are you? #Person1#: Not bad. Li Hong, I am sorry that I can't go to see Mrs. Brown with you tomorrow morning. My mother is ill. I must take care of her. #Person2#: I'm sorry to hear that. You'd better stay at home. After all, we can visit Mrs. Brown later. #Person1#: OK. Bye - bye. #Person2#: ...	<pad> Alice can't go to Mrs. Brown with Li Hong because Alice's mother is sick. Li Hong advises her to be at home to stay at home. </s>	<pad> Alice can't go to the doctor with Li Hong because Alice's mother is ill. Li Hong does not ask her to go to the doctor in the morning because she'll be home later. </s>	1.203251	1.370589	0.167338

Webapp

- Used Flask to create a small web app for the inference.
- Below is the interface of our app.



Deploy

- Dockerised our app and deployed using Hugging face space.
- Hugging Face provides a free space where we can deploy our small apps.
- [Github](#)
- [Hugging Face](#)

Conclusion

- The PEFT technique helps reduce computing power requirements.
- The performance of the PEFT model is comparable to a fully fine-tuned model.
- The RougeL score of the PEFT model reached 0.40, significantly higher than the 0.24 of the original model.
- After fine-tuning with reinforcement learning (RL), the toxicity of the model's generated output decreased by 52%.
- The overall model performance was suboptimal due to computational constraints.
- We were unable to experiment with hyperparameters extensively because training took too long and occasionally terminated unexpectedly.

Thank You